

7. An investigation was performed to discover the effect of light intensity on the stomatal density of the leaves of coffee plants (*Coffea arabica*). The drawings below were obtained using the high power objective of a light microscope. **Diagrams 1** and **2** show the number of stomata in the field of view on the lower epidermis for a leaf in full sun and a leaf in shade. Both diagrams are drawn to the same scale.

Diagram 1

Leaf growing in full sun

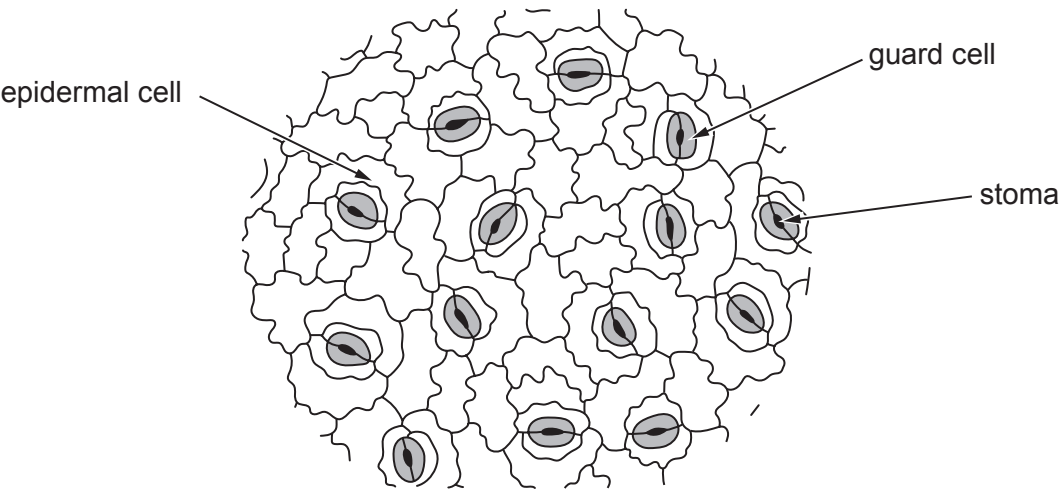
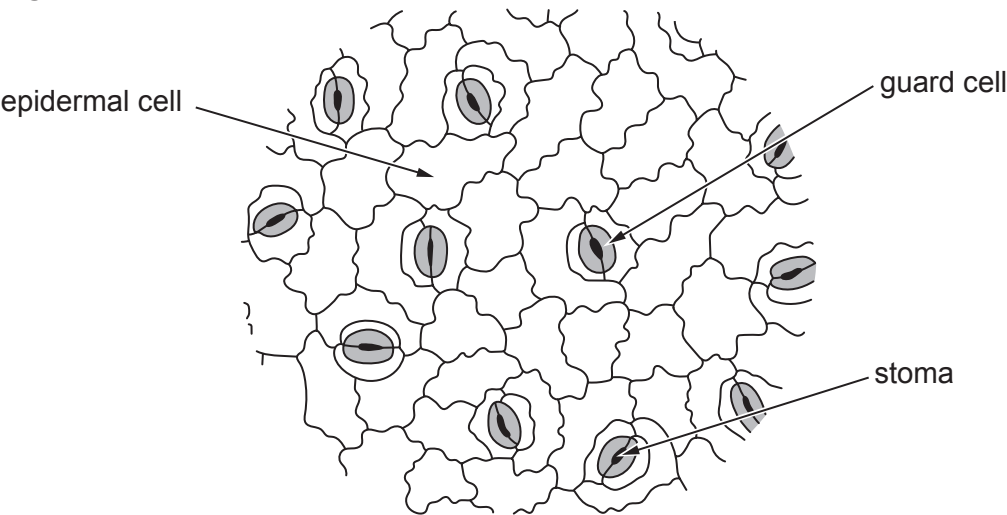


Diagram 2

Leaf growing in shade



- (a) With **the exception of shape**, state **two** structural features of guard cells that are different from epidermal cells. [2]

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- (b) (i) Use the following information to calculate the number of stomata per mm² on the leaf grown in full sun. Show your working in the space provided. **Give your answer to three significant figures.** [4]

Number of stomata = 14

Diameter of field of view = 0.3 mm

Area of a circle = πr^2

r = radius of a circle

π = 3.14

Number of stomata = stomata per mm²

- (ii) The stomatal index is the ratio of the number of stomata to the total number of epidermal cells. It is calculated using the following equation.

$$SI = \frac{S}{S + E} \times 100$$

SI = stomatal index.
S = number of stomata per unit area.
E = number of epidermal cells in the same unit area.

Table 1 gives the stomatal indices for the leaf grown in full sun and the leaf grown in shade.

Table 1

	Leaf grown in full sun	Leaf grown in shade
S	14	11
S + E	77	61
SI	18.2	18.0

